QP CODE: 25901752A SERIES: A

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCPHY100 - OBSERVATIONAL ASTRONOMY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour Maximum Marks: 35

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A

Multiple Choice Questions

Answer all questions
Each question carries 1 mark

1. According to the Copernican model, which celestial body is at the center of the universe?
   Ans: **C) Sun** [U] / [CO1]

2. Which unit is larger: 1 light year or 1 Astronomical Unit (AU) or 1 parsec ?
   Ans: **C) 1 parsec** [U] / [CO1]

3. What force keeps planets in orbit around the Sun?
   Ans: **B) Gravity** [U] / [CO1]

4. Which of the following constellations is known as “The Hunter”?
   Ans**: B) Orion** [K] / [CO4]

5. During which season do many trees shed their leaves?
   Ans: **C) Autumn** [K] / [CO4]

6. What kind of rays are used in radio astronomy?
   Ans: **C) Radio Waves** [K] / [CO2]

7. Which Space agency is monitoring James Webb Space Telescope?
   Ans: **A) NASA** [K] / [CO2]

8. What do refractive telescopes use to focus light?
   Ans: **B) Lenses** [K] / [CO2]

9. What is the approximate surface temperature of Sun?
   Ans: **C) 6,000 K** [K] / [CO5]

10. What is the approximate duration of totality in a solar eclipse ?
    Ans: **B) 7.5 mins** [K] / [CO5]

11. Which among the following is not a Jovian planet?
    Ans: **A) Mercury** [K] / [CO5]

12. Which type of galaxy has well-defined arms?
    Ans: **B) Spiral** [K] / [CO4]

13. The birth stage of a star is named as:
    Ans: **C) Protostar** [K] / [CO3]

14. Which spectral class is the Sun classified under?
    Ans: **B) G** [K] / [CO3]

15. Compared to a main sequence star, a red giant is:
    Ans: **B) Larger and Cooler** [U] / [CO3]

[1x15 = 15]

Part B

Multiple Choice Questions

Answer all questions
Each question carries 2 marks

16. What does the term "deferent" refer to in the Ptolemaic model?
    Ans**: C) The larger circle along which the center of the epicycle moves** [K] / [CO1]

17. Which book did Galileo publish to support the heliocentric theory?
    Ans: **A) Dialogue Concerning the Two Chief World Systems** [K] / [CO1]

18. Name the two points where the celestial equator and the ecliptic intersects.
    Ans: **B) Equinoxes** [K] / [CO4]

19. What do you mean by Right Ascension?
    Ans: **A) The celestial equivalent of Earth's Longitude** [K] / [CO4]

20. Hubble Space Telescope has primary mirror size of:
    Ans: **A) 2.4 meters** [K] / [CO2]

21. How are neutron stars detected from Earth?
Ans: **C) By their X-ray and radio wave emissions** [U] / [CO3]

22. What is the typical size and mass of a white dwarf?
Ans: **B) Same size as Earth with nearly the Sun's mass** [U] / [CO3]

23. How many stars are there approximately in Milky Way galaxy?
Ans: **A) 200 billion** [K] / [CO4]

24. Which celestial body most commonly causes occultations seen from Earth?
Ans: **A) Moon** [U] / [CO5]

25. What type of objects are found in the Kuiper Belt?
Ans: **B) Icy bodies, dwarf planets, and comets** [K] / [CO5]

[2x10 = 20]

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